Software Quality Engineering

Project Requirements

Students are required to work in groups of 2–3 members to design, develop, test, and evaluate a software application while applying the principles and practices of Software Quality Engineering (SQE).

Each group will identify a real-world problem and propose a software-based solution, such as an Online Examination System, Student Attendance Management System, Hospital Appointment System, Library Management System, E-Learning Platform, Inventory Management System, or another application approved by the instructor.

The primary objective is not merely to develop a functional application but to ensure that the software meets specified quality requirements, behaves reliably under different conditions, and is maintainable and usable.

Students must apply software quality engineering practices throughout the software development lifecycle, including requirements analysis, software design, implementation, software reviews, test planning, test-case design, test execution, defect tracking, and quality evaluation.

Each group must use a version control platform such as GitHub to maintain the source code, document contributions, track issues, and demonstrate collaborative development practices.

Project Objectives

By completing this project, students will be able to:

1. Identify functional and non-functional software requirements
2. Define measurable software quality attributes and acceptance criteria
3. Apply software review and quality assurance techniques
4. Design and execute test cases using appropriate testing techniques
5. Perform unit, integration, system, and user acceptance testing, where applicable
6. Identify, document, classify, and track software defects
7. Use software quality metrics to evaluate product quality
8. Apply version control and collaborative software development practices
9. Prepare professional software quality documentation and present evidence of testing and quality improvement

Project Scope

Each group must develop a small- to medium-sized software application with a clearly defined scope. The application should contain sufficient functionality to demonstrate meaningful testing and quality assurance activities.

Students are expected to identify quality risks, establish measurable quality criteria, conduct systematic testing, analyze the results, and recommend improvements based on the evidence collected.

The project will be evaluated on both the software product and the quality engineering process followed during its development.

Deliverables

Deliverable 1: Project Proposal

Initial submission

- Project title and problem statement

- Project objectives and scope

- Target users and key features

- Proposed technology stack

- Group members and assigned responsibilities

- Preliminary project schedule

Deliverable 2: Software Requirements Specification (SRS)

- Functional requirements

- Non-functional requirements, including usability, reliability, performance, and security where relevant

- Use-case diagram or user stories

- Acceptance criteria for important requirements

- Traceability between requirements and test cases

Deliverable 3: Software Design and GitHub Repository

- System architecture or component diagram
- Database design or data model, where applicable
- User interface prototypes or screenshots
- GitHub repository containing source code
- Meaningful commits, README, and issue tracking

Deliverable 4: Software Quality Assurance Plan

- Quality objectives and quality attributes
- Quality standards or coding conventions to be followed
- Review checklist for requirements, design, or code
- Risk identification and mitigation plan
- Testing strategy, responsibilities, and schedule

Deliverable 5: Test Plan and Test Cases

- Test scenarios and test cases with preconditions, inputs, expected results, and actual results
- Positive, negative, boundary-value, and equivalence-partitioning tests, as appropriate
- Unit and integration tests, where applicable
- System testing and user acceptance testing
- Requirements-to-test traceability matrix

Deliverable 6: Test Execution and Defect Report

- Test execution results with supporting evidence
- List of passed and failed test cases
- Defect log containing defect ID, description, severity, priority, status, and reproduction steps
- Evidence of defect correction and retesting, where applicable
- Summary of unresolved defects and known limitations

Deliverable 7: Software Quality Metrics Report

Students must calculate and interpret at least three relevant metrics, such as:

- Test pass rate

- Requirements test coverage

- Defect density, if a meaningful size measure is available

- Defect removal or retesting effectiveness, if sufficient data is collected

- Response time or task completion rate, where relevant

Include calculations, collected evidence, interpretation, and recommendations. Do not report invented or unsupported measurements.

Deliverable 8: Final Software Product and Report

- Working application or demonstrable prototype

- Complete source code and setup instructions

- Final report consolidating requirements, design, QA activities, testing, defects, metrics, and limitations

- Screenshots and other evidence of testing

- Individual contributions of group members

- Future improvement recommendations

Deliverable 9: Final Demonstration and Viva

- Project presentation

- Live demonstration of the application

- Demonstration of selected test cases and defect handling

- Explanation of quality metrics and results

- Individual viva to assess each student's understanding and contribution